

Math 8 — Q1 Mid-Quarter Test

Mid-Q1 Content · 15 questions

Name: _____ Class: _____ Date: _____

Directions: Show all work. Box or circle your final answer. Include units where appropriate. No calculators. Write exponents with the caret notation ($10^7 = 10$ to the 7th power).

Metric Conversion Reference — KHDUdcm

Kilo	Hecto	Deca	Unit (m/g/L)	deci	centi	milli
k	h	da	m (meter) / g (gram) / L (Liter)	d	c	m

Going **UP** the ladder (smaller → larger unit): move decimal **LEFT** one place per step. Going **DOWN** (larger → smaller): move decimal **RIGHT** one place per step.

- Convert 3.2 km to millimeters. Answer: _____
- Convert 45 cm to meters. Answer: _____
- A student measured 24.5 g; actual mass is 25 g. What is the percent error? Answer: _____
- Compute: $-15 + 8$. Answer: _____
- Compute: $(-4) \times (-7)$. Answer: _____
- Compute: $(-12) - (-4)$. Answer: _____
- A blue whale weighs 180,000 kg (actual). A scientist measured it as 175,000 kg — an error of 5,000 kg. A house cat weighs 8 kg (actual). Another scientist measured it as 7 kg — an error of 1 kg. Which measurement is MORE accurate? Answer: _____
- Write 82,500,000 in scientific notation. Answer: _____
- Write 0.00051 in scientific notation. Answer: _____
- Compute: $(4 \times 10^5)(3 \times 10^2)$. Give in scientific notation. Answer: _____
- Compute: $(9 \times 10^8) \div (3 \times 10^3)$. Give in scientific notation. Answer: _____
- Convert 3.2×10^{-2} to standard form. Answer: _____
- Simplify (Laws of Exponents): $3^4 \cdot 3^2$. Answer: _____
- Simplify: $(7^5) \div (7^2)$. Answer: _____
- Simplify: $(2^3)^4$. Answer: _____